

Dagem Dereje

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SUMMARY

Recent B.Sc. graduate in Data Science with strong foundations in machine learning, deep learning, computer vision, statistical modeling, and applied AI research. Completed a research-focused internship at the Information Network Security Administration (INSA), where I conducted literature reviews on emerging AI technologies, contributed to deep learning research projects, and implemented a human pose estimation system using modern computer vision techniques.

Recipient of the Best Senior Project Award across all departments in the College of Science, Bahir Dar University, for developing an AI-based drought prediction system that applies machine learning and deep learning techniques to forecast meteorological drought using historical climate data from the Ethiopian Meteorological Institute (EMI). Passionate about transforming research into practical AI solutions for real-world challenges.

EDUCATION

Bahir Dar University

Bachelor of Science (B.Sc.) in Data Science, CGPA: 3.52 / 4.00, National Exit Examination: 84 / 100, Best Senior Project Award – College of Science (Selected as the best capstone project across all departments), Certificate of Recognition for Outstanding Senior Project

EXPERIENCE

Data Science Research Intern

Information Network Security Administration (INSA)

Addis Ababa, Ethiopia

- Conducted comprehensive literature reviews on Deepfake Generation, Deepfake Detection, Text-to-Speech (TTS), and Human Pose Estimation.
- Studied and evaluated modern deep learning architectures including HRNet, Lite-HRNet, Faster R-CNN, and other computer vision models.
- Prepared and presented technical presentations summarizing recent AI research papers followed by technical discussions with mentors.
- Contributed to implementing a Human Pose Estimation system capable of detecting and connecting 17 human body keypoints using the COCO dataset.
- Debugged and modified PyTorch-based project code to support customized keypoint detection and multi-person pose estimation.
- Integrated Faster R-CNN into the pose estimation pipeline for multi-person detection.
- Trained and validated deep learning models using GPU environments on Kaggle and Google Colab before large-scale training on INSA GPU infrastructure.
- Documented experimental results, model performance, and research findings throughout the project.
- Technologies: Python • PyTorch • TensorFlow • OpenCV • NumPy • COCO Dataset • Google Colab • Kaggle • PyCharm

PROJECTS

AI-Based Drought Prediction System (Award-Winning Senior Project) | Developed an AI-driven drought forecasting system that predicts future meteorological drought (SPI-3) using historical climate data obtained from the Ethiopian Meteorological Institute (EMI). The project applied machine learning and deep learning models to support early drought prediction and evidence-based decision-making for disaster preparedness and agricultural planning.

- Technologies: Python, Scikit-learn, TensorFlow, Pandas, NumPy, XGBoost, Deep Learning

Human Pose Estimation using Deep Learning | Implemented a computer vision system capable of detecting 17 human body keypoints by integrating Faster R-CNN with HRNet and Lite-HRNet architectures. Modified and debugged the deep learning pipeline, trained the model using GPU resources, and evaluated pose estimation performance using the COCO dataset.

- Technologies: Python, PyTorch, OpenCV, HRNet, Lite-HRNet, Faster R-CNN, COCO Dataset

Humanitarian Aid Resource Allocation & Decision Support System | Developed an AI-driven decision support system to optimize humanitarian aid distribution using data analytics and machine learning. Performed data collection, preprocessing, exploratory data analysis, feature engineering, predictive modeling, and dashboard development to support evidence-based resource allocation and improve the efficiency and transparency of humanitarian operations.

• Technologies: Python, Scikit-learn, Pandas, NumPy, SQL, Power BI, Excel

Programming: Python, SQL, Java, JavaScript, HTML, CSS

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, CatBoost

Computer Vision: OpenCV, HRNet, Lite-HRNet, Faster R-CNN

Data Science: Pandas, NumPy, Feature Engineering, Data Cleaning, Statistical Analysis, Predictive Modeling, Model Evaluation

Development Tools: Jupyter Notebook, Google Colab, Kaggle, PyCharm, VS Code, Git, Git hub

RESEARCH EXPERIENCE

- Artificial Intelligence Research - Machine Learning - Deep Learning - Computer Vision - Deepfake Generation and Detection - Human Pose Estimation - Text-to-Speech Technologies - Scientific Literature Review - Technical Paper Analysis - Model Evaluation and Comparative Analysis

ACHIEVEMENTS

- Best Senior Project Award across all departments in the College of Science, Bahir Dar University. - Successfully completed a research-focused internship at INSA involving advanced AI and computer vision research. - Conducted and presented technical reviews of state-of-the-art deep learning research papers. - Developed machine learning solutions addressing real-world challenges in drought prediction, computer vision, and fraud detection.

LANGUAGES

Amharic – Native

English – Professional Working Proficiency